AVIATION SYSTEMS ENGINEERING · RESEARCH CONCEPT REPORT

ANIRBAN-1: IoT Mesh for Autonomous Aircraft Safety

Integrated Technical Concept Report — Cooperative Trajectory Sharing, Conflict Prediction, and Advisory-Layer Engineering

Contributors: Abu Kayes Huda, Jaidul Islam Alvi, MD Sakib Hasan

Report date	Revision	Document type	System boundary
15 October 2026	1.0	Concept / research architecture	Advisory-only, simulation-first

> **Status: Research Concept — Not for Operational Use.** This document is a research architecture and concept report. It is not an installation manual, a certified design, an operational flight procedure, or evidence of compliance with any standard named in it. Naming a standard does not establish compliance. Any real-aircraft modification must be designed and approved by qualified aviation specialists and the responsible authority.

Contents

1 Document Status and Scope of This Report
2 Executive Summary
3 Introduction, Motivation, Problem Statement, and Research Question
4 System Scope, Objectives, Assumptions, and Non-Goals
5 System Architecture and Requirements
6 Surveillance Inputs, Sensors, and Non-Cooperative Object Detection
7 Communications, Message Formats, Coordination, and Failure Behaviour
8 Onboard Software Modules and Processing Workflow
9 Human Factors and the Advisory Interface
10 Preliminary Hazard Analysis, Safety Considerations, and Cybersecurity
11 Applicable Standards and Potential Certification Pathway
12 Validation Methodology, Baselines, Metrics, and Acceptance Criteria
13 Development Roadmap and Evidence Gates
14 Discussion, Research Gap, Proposed Contributions, and Limitations
15 Decisions Made During Revision of the Original Concept
16 Conclusions and Next Steps
17 References

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

18 Glossary

19 Contributor Statements

Figures are numbered 1–10 and Tables 1–25 in document order. Each figure is placed beside the argument it supports and is cited explicitly in the text.

1 Document Status and Scope of This Report

This is revision 1.0 of the ANIRBAN-1 concept report. It integrates the technical substance of the supplied concept report and its accompanying research-and-technology comparison and system-architecture design into one document, and it revises the presentation of the original concept where the original overstates evidence.

Two source representations were inspected: the report HTML (primary editable source) and its rendered PDF (quality-assurance reference). They are different renderings of the same study, not two separate studies. The PDF inspection identified the defects corrected here: an opening chart gallery detached from the narrative, tables that collapsed into single merged cells and misaligned headers on export, a duplicated glossary and contributor block at the end of the document, substantial blank page area, and a radar update interval drawn as a single fixed value at the top of a reported 4–12 second range.

Every visualization in this edition is hand-authored inline SVG. No remote chart or diagram script is loaded, so figures cannot render blank or malformed during print or PDF export. Sections 2–19 carry the technical content; Section 15 records what was changed relative to the original concept and why.

> **Engineering boundary maintained throughout:** ANIRBAN-1 is a simulation-first, advisory-only research concept. It does not command flight controls, and it does not replace, disable, or contradict existing TCAS/ACAS, ADS-B, ATC, or established operating procedures. No claim is made that the proposed system is safer, more accurate, more reliable, or more effective than existing aviation systems.

2 Executive Summary

ANIRBAN-1 proposes a cooperative, advisory-only aircraft safety overlay. Participating aircraft would exchange state and planned-trajectory intent, predict potential conflicts over a declared horizon, evaluate candidate resolutions against explicit constraints, and present an explainable recommendation to a pilot or operator. Existing collision-avoidance and air-traffic functions remain independent and unmodified throughout.

The founding research question is answerable, and the answer is currently "unproven but testable." The question — whether a carefully engineered, fault-aware cooperative trajectory-sharing and conflict-prediction layer can demonstrate measurable benefits beyond existing surveillance, collision-avoidance and trajectory-management approaches — can be tested on bench computers and simulators at low cost. However, no public evidence establishes that such a layer already exceeds the ACAS X / DAIDALUS / trajectory-based-operations class of systems. The defensible research frontier is narrower than the original concept assumed: not smarter avoidance logic, and not a new network, but whether aircraft can share intent and coordinate under fault, uncertainty and disagreement without ever making their existing safety depend on that cooperation.

Six corrections to the original concept pitch are carried through every subsequent section:

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

1. **TCAS is not a last-second panic button.** ACAS II / TCAS II operates independently of ground ATC and issues Traffic and vertical Resolution Advisories. It is a complement ANIRBAN-1 must never contradict.
2. **Ground radar is not obsolete.** ADS-B complements surveillance with roughly one-second GNSS-derived broadcasts, but visibility depends on equipage, coverage and reception, and the broadcast carries no native authentication.
3. **6G is not an available aviation safety network.** IMT-2030 performance requirements were drafted in February 2026 with commercial framing around 2030. A dedicated 6G air-to-air mesh is future research, not a prototype dependency.
4. **Consensus and ledgers cannot establish sensor truth.** Authentication proves message origin and integrity only. General-purpose blockchain consensus is removed from the time-critical loop; coordination uses a deterministic, timeout-bounded proposal/acknowledgement protocol with declared disagreement outcomes.
5. **LiDAR, cameras and radar are not universal detectors.** Airborne small-target detection is a bounded research question requiring measured probability-of-detection and false-alarm characteristics against a declared target envelope.
6. **The mesh itself is not the invention.** IEEE 1920.1 already defines infrastructure-free air-to-air ad hoc networking. ANIRBAN-1 should build on such services and position its contribution above the transport layer.

The recommended Version 0.1 is defined by ten system requirements (SYS-001 to SYS-010, Table 4), two of which are hard gates: no flight-control output (SYS-007) and fail-closed suppression of advice when inputs are inadequate (SYS-006). Validation proceeds through stages V0–V6 with declared exit evidence, benchmarked against three baselines and using ablation runs that disable one feature at a time so that any measured improvement is attributed rather than asserted. Novelty is stated strictly as four hypotheses (C1–C4, Table 22), each paired with the evidence it would require.

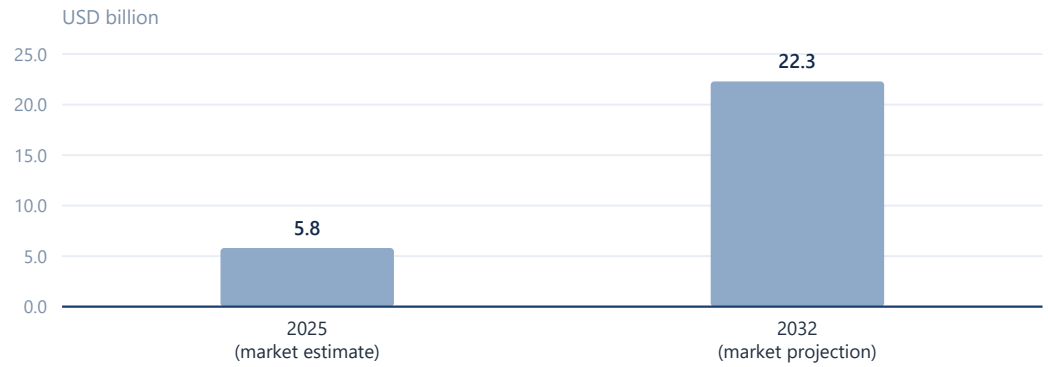
Numbers that could not be verified in the supplied sources — per-aircraft cost bands, power draw, fuel-burn impact, communication uptime targets, alert lead-time figures, certification timelines, and assigned assurance levels — are recorded as unverified or removed rather than replaced with invented estimates. Section 15 lists every such decision.

3 Introduction, Motivation, Problem Statement, and Research Question

3.1 Motivation and industry context

The motivation for a cooperative layer is that tactical separation today is delivered by systems designed for a different problem. Onboard collision avoidance acts close to an encounter and normally constrains vertical manoeuvres; surveillance broadcast publishes current state, not a negotiated future path; and air-traffic services coordinate strategically through ground-mediated processes. A peer-to-peer exchange of intent is the missing element, and the commercial aviation connectivity market is often cited as evidence that the enabling infrastructure is maturing. Figure 1 shows that market framing, and the caveat matters more than the curve.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0



Third-party market estimate and projection (21.2% CAGR). Market size is not a technical validation metric.

Figure 1. Global aviation IoT market: reported estimate and projection

Third-party market research reports the aviation IoT market at USD 5.8 billion in 2025 (estimate) with a projection of USD 22.3 billion by 2032, a compound annual growth rate of 21.2%. Both bars are third-party figures, not measurements, and the two points are separated by seven years of forecast uncertainty.

Source: Market research report cited in the supplied research-and-technology comparison (accessed 15 October 2026). Reference 33 in Section 17.

> **Interpretation limit.** Market size is not a technical validation metric. The reported growth concentrates in predictive maintenance, asset tracking and cabin services; the same industry surveys identify cybersecurity, functional safety and sector conservatism as the three adoption challenges. A safety-adjacent collision function sits at the hardest certification tier of exactly those three challenges. Figure 1 therefore supports a statement about connectivity maturity only, and is retained here for context rather than as evidence that ANIRBAN-1 would work.

3.2 What existing collision avoidance already achieves

Any claim about an additional safety layer must be measured against the systems that already exist. The most important reference point is the successor to TCAS II. The ACAS X development line replaces fixed rule tables with Markov-decision-process logic and machine-learning-optimized lookup tables, supports vertical, horizontal and blended resolution advisories, and fuses ADS-B In data. Programme-reported results attribute a 65 per cent reduction in unnecessary alerts and a 20 per cent reduction in mid-air collision risk relative to TCAS II, as shown in Figure 2.

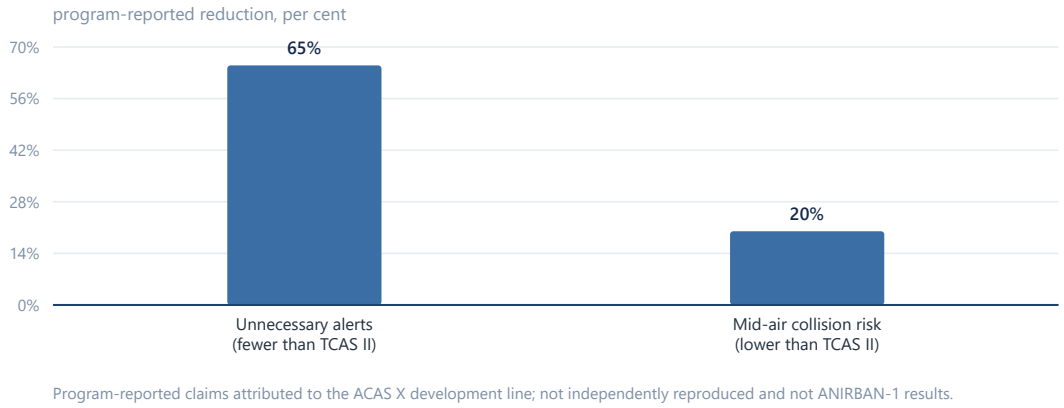


Figure 2. ACAS X programme-reported improvements relative to TCAS II

Two programme-reported reductions: 65% fewer unnecessary alerts and 20% lower mid-air collision risk. These are claims attributed to the ACAS X development line. They have not been independently reproduced in this report, and they are not ANIRBAN-1 results.

Source: Programme data summarized in the supplied research-and-technology comparison; corroborated by the ACAS X programme overview. References 3 and 19 in Section 17.

The implication for ANIRBAN-1 is uncomfortable but clarifying. Optimized, sensor-flexible, probabilistic collision avoidance with multi-directional advisories is a funded, standardized, legislatively advancing programme: European legislation already accepts configurations requiring aircraft above defined seat or weight thresholds to equip ACAS Xa or ACAS II 7.1. ANIRBAN-1 novelty therefore cannot rest on smarter avoidance logic. What remains open is the peer-to-peer fault semantics above it — and Figure 2 sets the bar against which any ANIRBAN-1 alerting claim must be compared, not surpassed by assertion.

3.3 Related technology: trajectory-based operations

Sharing a future path is not itself unprecedented. Trajectory-based operations (TBO) share and dynamically maintain a four-dimensional trajectory among air traffic control, airlines and aircraft, with the flight management system managing speed against required arrival times. National programmes have moved this from paper to flight. Figure 3 reports the contents of one such verification.

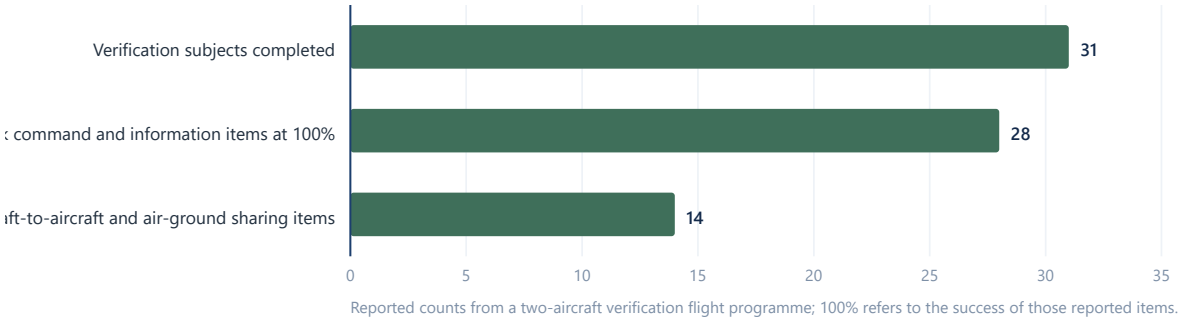


Figure 3. Reported contents of a two-aircraft trajectory-based-operations verification

Reported counts from a completed two-aircraft verification: 31 verification subjects, 28 datalink command and information items, and 14 aircraft-to-aircraft and air-ground situation-sharing items, all reported at 100% success. The 100% figure refers to the success of those reported items in that programme.

Source: Civil aviation authority news report of a completed two-aircraft TBO verification (December 2024), cited in the supplied research comparison. Reference 24 in Section 17.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Figure 3 establishes that minutes-horizon trajectory sharing between aircraft and ground services is being implemented by state programmes. It does not establish anything about ANIRBAN-1. A two-aircraft verification does not demonstrate safety case adequacy, scalability to dense traffic, certification readiness, or the effectiveness of a different architecture. Its relevance to this report is narrow and specific: it confirms that the intent-sharing premise rests on live groundwork, which shifts ANIRBAN-1 contribution away from the trajectory mathematics and toward what those programmes do not yet supply — aircraft-to-aircraft intent exchange with declared fault semantics between peers, without waiting for ground-mediated agreement.

3.4 Problem statement and research question

The problem ANIRBAN-1 addresses is therefore not the absence of collision avoidance, surveillance or trajectory management. It is the absence of a documented, testable specification for *peer coordination under fault*: what an aircraft does when a peer's intent message is stale, missing, contradictory, expired, or adversarial; what happens when the network partitions; and how a human is told that the system no longer knows enough to advise.

The research question is stated for testing, not for advocacy:

> **Research question.** Can a carefully engineered, fault-aware cooperative trajectory-sharing and conflict-prediction layer demonstrate measurable benefits beyond existing aviation surveillance, collision avoidance, and trajectory-management approaches — where "benefits" are defined by pre-declared metrics, measured against named baselines, and attributed by ablation?

This report answers the question honestly: at concept stage the answer is that the question is testable and the testbed is inexpensive, and that the benefit remains undemonstrated. Sections 12 and 14 define what would constitute an acceptable answer.

4 System Scope, Objectives, Assumptions, and Non-Goals

4.1 Version 0.1 scope

Version 0.1 operates primarily in simulation. It processes simulated or appropriately available aircraft-state and traffic information; maintains a local representation of nearby traffic with declared uncertainty; predicts potential conflicts over a declared prediction horizon; evaluates alternative trajectories under explicit constraints; exchanges simulated aircraft intent and candidate proposals; handles stale messages, communication loss, disagreement and invalid inputs; presents recommendations through a research interface; and remains isolated from flight-control commands, leaving existing ACAS/TCAS functions and applicable operational procedures independent.

It does not begin by assuming that aircraft automatically execute maneuvers negotiated by an algorithm. Expansion paths — hardware-in-the-loop testing, possible approved aircraft integration, additional sensors, new communication services, eventual certification — are described separately in Section 13, each with its own prerequisites and exit criteria.

Table 1. Version 0.1 design decisions and rationale

Decision area	Baseline choice	Reason

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Decision area	Baseline choice	Reason
Product boundary	Advisory-only cooperative conflict predictor	Meaningful testing without controlling an aircraft
Initial traffic data	Simulated tracks; recorded data where permitted	Repeatable tests before any real aircraft interface
Compute	Research workstation / bench computer	Avoids prematurely selecting flight-qualified hardware
Network	Virtual network; lab network only under authorization	Tests intent messages, delay, loss and partitions safely
Coordination	Deterministic proposal/acknowledgement with timeouts	Simpler to test than consensus; explicit failure behaviour
Sensors	Optional module, evaluated separately	Camera/LiDAR/radar performance needs target- and environment-specific evidence
Display	Conventional advisory panel first	Human-factors evidence before any AR or HUD integration
Aircraft integration	None until aircraft and approval path are defined	Placement, buses, power, antennas and approvals are airframe-specific

Consolidated from the supplied system-architecture design document (Rev 0.1). These choices replace the original concept report's framing, in which a negotiated trajectory was fed to the flight-control system and the pilot's default was to follow it automatically.

4.2 Explicit non-goals

Six non-goals define the evidentiary boundary of Version 0.1. Each removes a burden of proof the programme cannot currently meet, and each is paired with a future expansion path that carries its own prerequisites (Section 13).

- Replace TCAS/ACAS, ADS-B, radar, ATC, the flight management system, or operating procedures.
- Automatically steer, climb, descend, slow, or send commands to an autopilot or flight-control computer.
- Assume all aircraft carry ANIRBAN equipment or share complete future four-dimensional trajectories.
- Claim that LiDAR or cameras see all birds, drones, balloons, gliders, or non-cooperative aircraft.
- Require live 6G, cloud connectivity, or blockchain to detect a local conflict.
- Claim certification compliance on the basis of this concept document.

The second non-goal is decisive. Once the system has no write path to flight control, its worst failure class changes from an aircraft-level safety event to misleading information, which is what permits bench-computer development at research pace. The sixth pre-empts the most common failure mode of concept documents of this kind: presenting a list of standard names as if it were an assurance argument.

4.3 Standing assumptions

- **Unequipped traffic exists.** A portion of nearby traffic will never participate; the local picture must remain usable with cooperative data only.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

- **Intent is a claim.** A received intended trajectory is a statement about planned behaviour, not a guarantee of future position.
- **Networks fail.** Loss, delay, reordering, duplication and partition are normal conditions to be designed for, not anomalies.
- **Authentication is not truth.** A correctly signed message may still carry an incorrect navigation solution, a wrong clock, or a compromised sender.
- **Access is permission-bound.** Direct access to flight-management or collision-avoidance internals is aircraft-specific and must never be assumed.
- **A criterion must be declared.** Every conflict result depends on the chosen separation or well-clear criterion and its operating domain.

5 System Architecture and Requirements

5.1 Layered architecture

The architecture separates external data acquisition, local airspace state, prediction, planning and coordination, an independent proposal check, the human interface, and evidence collection. Its organising principle is evidential hygiene: nothing reaches the prediction path without a timestamp, a source identity, a validity flag, and an uncertainty representation or an explicitly declared unknown. Figure 4 shows the layers and their boundaries.

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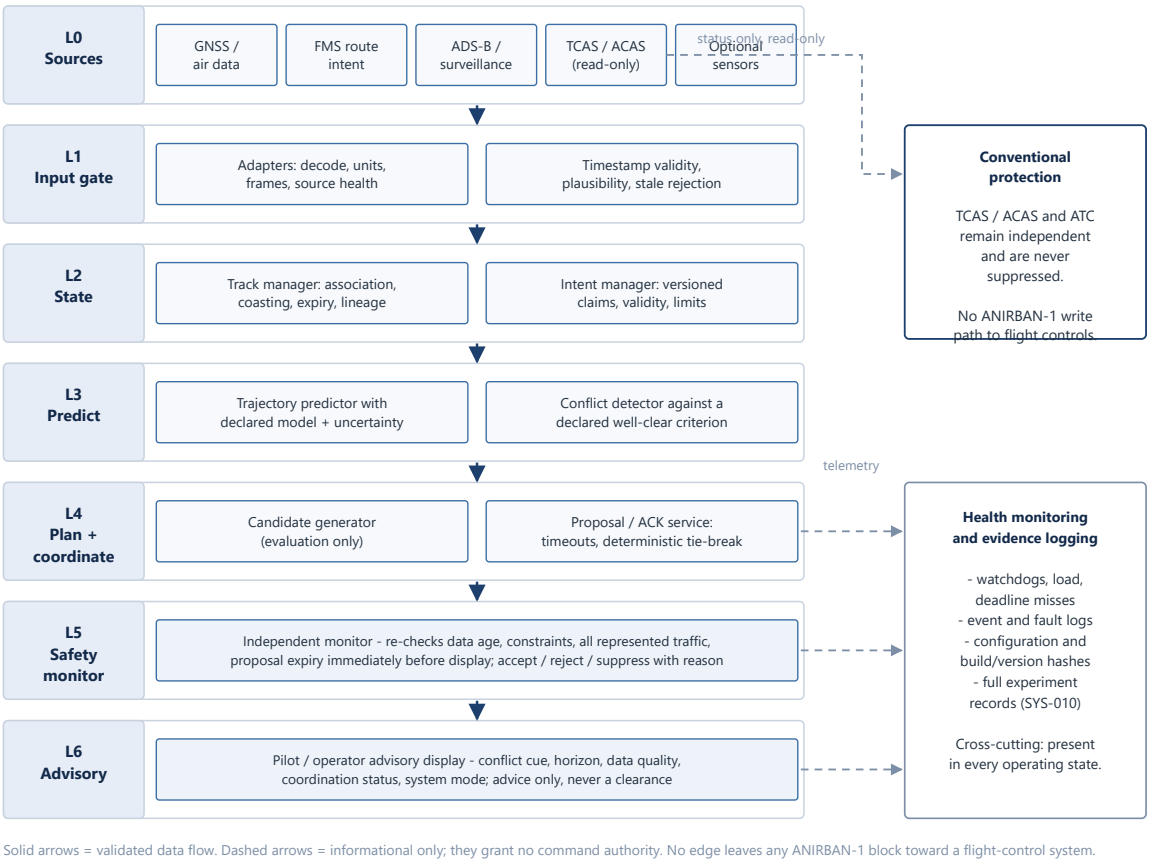


Figure 4. ANIRBAN-1 Version 0.1 layered system architecture (L0–L6)

Data flows downward through seven logical layers from aircraft and scenario sources (L0) to the advisory display (L6). The independent safety monitor (L5) re-checks every candidate immediately before display. Conventional protection (TCAS/ACAS and ATC) and the health/evidence function are drawn outside the main data path: dashed links are informational and grant no command authority. No edge leaves any ANIRBAN-1 block toward a flight-control system.

Source: Engineering design consolidated from the supplied system-architecture design document (Rev 0.1, 15 October 2026) and extended for this revision.

Figure 4 illustrates two properties that are architectural rather than stylistic. First, the independent monitor sits between planning and display, so an unchecked candidate cannot reach the pilot by passing through the planner alone. Second, conventional protection is drawn as a separate boundary that the advisory layer may read but never suppress. Together these encode requirement SYS-007 and hazard HZ-06 structurally.

Table 2. Logical layers L0–L6 and cross-cutting functions

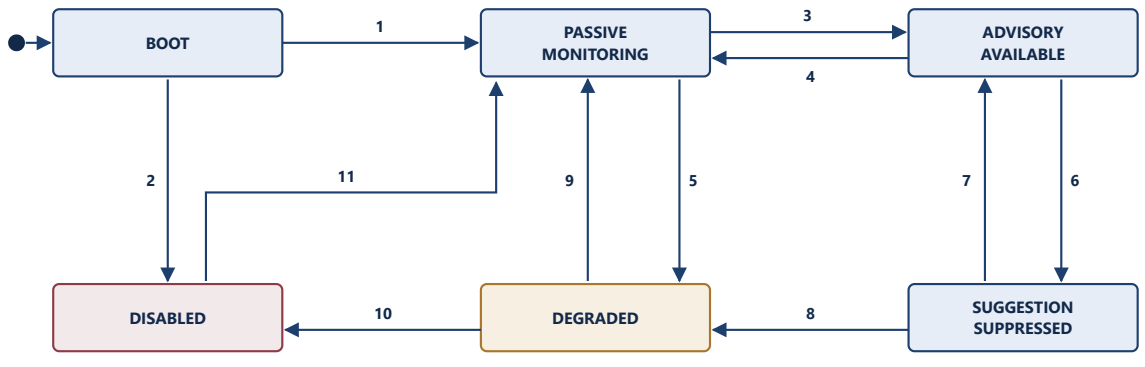
Layer	Main components	Primary outputs
L0 Aircraft / scenario sources	Navigation and air data; route intent where available; traffic surveillance (ADS-B derived); optional sensors; ATC/ACAS state only where properly provided	Raw observations with source, time and quality
L1 Input boundary	Source adapters; unit normalization; timestamp checks; validity rules; source health	Validated observation events, or explicit invalid-data states

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Layer	Main components	Primary outputs
L2 State estimation	Track manager (create, associate, coast, expire, source lineage); intent manager (versioned claims); ownship state; uncertainty and data age	Versioned local airspace state
L3 Prediction	Trajectory model with declared assumptions and uncertainty propagation; conflict detector against a declared criterion	Predicted conflict intervals with reason codes
L4 Planning / coordination	Candidate generator; multi-traffic re-check; proposal and acknowledgement exchange; deterministic tie-break and timeout logic	Candidate advisory proposal, or an explicit no-safe-recommendation state
L5 Safety monitor	Independent re-check of input validity, constraints, all represented traffic and proposal expiry	Accept, reject or suppress, with reason code
L6 Human interface	Advisory display; path visualization; confidence and data-age banner; acknowledgement	Pilot or operator information only
Cross-cutting	Health monitoring; logging; cybersecurity; configuration and version control; test harness	Traceable evidence and diagnosis

5.2 Operating states

Layer L5 and the advisory manager together enforce a small, closed set of operating states. The requirement is that degradation is always visible and always named: an aircraft's safety-relevant awareness moves through declared states rather than silently losing confidence. Figure 5 gives the state machine; Table 3 defines its transitions.



Numbered transitions are defined in Table 3. Every coordination outcome resolves to one of these declared states; no transition depends silently on reaching a peer.

Figure 5. Operating-state machine for the advisory-only prototype

Six declared states: BOOT, PASSIVE_MONITORING, ADVISORY_AVAILABLE, DEGRADED, SUGGESTION_SUPPRESSED and DISABLED. Eleven numbered transitions govern entry and exit. Transitions into DEGRADED and DISABLED are triggered by input or timing faults; SUGGESTION_SUPPRESSED is entered when the independent monitor rejects a candidate or a proposal expires.

Source: Derived from the supplied system-architecture design document (Rev 0.1) and the failure-handling table reproduced as Table 3.

Table 3. State transitions and triggers

#	From	To	Trigger	Required logging

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

#	From	To	Trigger	Required logging
1	BOOT	PASSIVE_MONITORING	Self-test passed and inputs valid	Build/config hash, self-test result
2	BOOT	DISABLED	Self-test failed or unauthorized configuration	Failure reason, configuration identity
3	PASSIVE_MONITORING	ADVISORY_AVAILABLE	Conflict predicted and candidate accepted by the monitor	Conflict set, candidate identifier, monitor decision
4	ADVISORY_AVAILABLE	PASSIVE_MONITORING	Conflict resolved below the declared criterion	Resolution time, criterion tag
5	PASSIVE_MONITORING	DEGRADED	Stale source, timing loss or deadline miss	Age, threshold from config, affected outputs
6	ADVISORY_AVAILABLE	SUGGESTION_SUPPRESSED	Monitor rejects the candidate or the proposal expires	Failed checks, rule identifiers, expiry
7	SUGGESTION_SUPPRESSED	ADVISORY_AVAILABLE	A fresh candidate is accepted	New candidate identifier, monitor decision
8	SUGGESTION_SUPPRESSED	DEGRADED	Essential inputs become inadequate	Input status, suppression reason
9	ADVISORY_AVAILABLE	DEGRADED	Peer unconfirmed where policy requires visibility	Timeout, peer identity, retry count
10	DEGRADED	DISABLED	Persistent fault or operator disable	Fault latch, operator action
11	DEGRADED / DISABLED	PASSIVE_MONITORING	Inputs recovered and revalidated, or manual re-enable with successful self-test	Recovery event, revalidation evidence

Every coordination outcome resolves to one of these states. No transition may depend silently on reaching a peer aircraft; local conflict detection and existing aircraft safety functions remain available in every state.

5.3 System requirements

Ten system-level requirements convert the concept into a testable specification. Each carries declared verification evidence, so that requirement satisfaction is an inspectable fact rather than an intention.

Table 4. Starter system requirements SYS-001 to SYS-010

ID	Requirement statement	Verification evidence
SYS-001	Accept timestamped ownship and traffic state records from a simulator or authorized data adapter	Interface tests; malformed-data tests
SYS-002	Every track carries source, age, quality status and an uncertainty representation or a declared unknown status	Unit tests; state-store inspection

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

ID	Requirement statement	Verification evidence
SYS-003	The predictor produces future-position estimates over a configurable study horizon	Ground-truth simulation comparison
SYS-004	The conflict detector evaluates tracks against a declared criterion selected for the study	Test vectors; baseline comparison
SYS-005	Every proposed alternative is checked against all represented traffic and stated constraints before display	Scenario tests; infeasible-candidate rejection
SYS-006	No valid recommendation is shown if essential inputs are stale, inconsistent or insufficient	Fault-injection tests; user-interface state tests
SYS-007	The prototype outputs no flight-control commands and does not modify TCAS/ACAS behaviour	Interface audit; code and architecture review
SYS-008	Loss, delay, duplication, replay-like input and network partition are observable and logged	Network fault-injection report
SYS-009	Each advisory states the conflict cue, prediction horizon, data-quality indicator, and candidate check status	Human-factors review
SYS-010	Every experiment records scenario configuration, algorithm version, inputs, outputs and metrics	Reproducibility audit

Retained unchanged from the supplied architecture document. SYS-007 must be verified by architecture and code audit so that the absence of a control path is structurally provable, not merely unexercised in tests. SYS-006 enforces fail-closed behaviour: an inadequate traffic picture produces a suppressed advisory and a visible degraded indication, never a confidently rendered path built from stale data.

5.4 Hardware architecture options

Three hardware architectures were compared. Cost and complexity are declared as qualitative categories; no equipment prices are stated, because the supplied documents contain no verified quotations.

Table 5. Hardware architecture options

Option	Composition	Cost / complexity	Stage	Prerequisites to enter the stage
H1 — Single research computer	One bench workstation running all modules in separated processes; simulated I/O; browser advisory panel	Low	Phase 1 (simulation MVP) — **recommended start**	None beyond software
H2 — Modular research island	Compute host plus isolated input-adapter unit, separate monitoring and watchdog unit, logging store; optional lab radios under authorization	Medium	Phase 3 (HIL prototype)	Stable module interfaces; timing budget measured in H1; authorization for any radio-frequency transmission

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Option	Composition	Cost / complexity	Stage	Prerequisites to enter the stage
H3 — Fault-tolerant airborne-class	Dual cross-monitored processing channels, qualified power and cooling provisions, EMC-hardened enclosures, approved interface gateway, airframe-specific mounting	High	Post-feasibility-study only (Phase 4 and beyond)	Selected aircraft and intended function; authority dialogue; certification basis; ARP4754-class development process established

H1 is the recommended starting point. Its one substantive hardware discipline is process isolation from the outset: the independent monitor runs as a separate process with its own watchdog, so the assurance argument being built in software anticipates the physical separation that H2 and H3 would require.

Interfaces, and what is deliberately not specified. ARINC 429 and ARINC 664/AFDX are plausible future carriers on specific aircraft types, but neither is a universal plug-in interface. Any Interface Control Document must define data definitions, units, label or message sets, update rates, timestamps, quality flags, access rights and failure behaviour before hardware decisions are taken. This report does not specify connector pinouts, antenna placement, bus loading, wire gauges, power budgets, cooling flows or part numbers. The original concept report's installation narrative — top and bottom fuselage antennas, nose and belly sensor fairings with heated windows, avionics-bay compute with cabin ventilation — is demoted to illustrative description, because those are exactly the airframe-specific decisions reserved for a manufacturer-authorized design study. A laboratory prototype and a qualified airborne installation remain distinct categories.

Table 6. Hardware element register: purpose, prototype status, and open airborne-design questions

Element	Purpose	Version 0.1 implementation	Open questions for any airborne installation
Compute host	Runs adapters, track store, prediction, monitor and interface service	Bench computer	Environmental qualification, power draw, cooling, partitioning, worst-case timing, maintainability
Input adapter / isolation	Converts simulator or authorized source records into a stable internal schema	Mock interfaces, files, sockets	Permitted aircraft interface, electrical isolation, bus loading, data-access authority
Communication gateway	Serializes, validates, authenticates where applicable, expires and routes research messages	Virtual network	Approved spectrum, airworthiness, interference assessment, range, network service, certification evidence
Optional sensor module	Turns recorded or simulated detections into bounded target tracks	Off, or recorded-data input	Target envelope, mounting, field of view, environment, EMC, aerodynamic and maintenance impact
Operator display	Shows advisory state, confidence, reason and recommended path	Browser or desktop panel	Cockpit integration, alert priority, visibility, latency, workload, training, approval
Health and storage	Collects state transitions, inputs, algorithm versions and test outputs	Local files and test database	Secure maintenance access, capacity, retention, privacy and security

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Element	Purpose	Version 0.1 implementation	Open questions for any airborne installation
Power, cooling and mechanics	Supports all installed equipment	Bench power supply	Load analysis, circuit protection, heat, vibration, fire zones, structural mounting, lightning and EMI

Fault containment follows the isolated-equipment-island principle: a malfunctioning experimental unit must not affect power quality, bus traffic, or display channels used by certified systems.

6 Surveillance Inputs, Sensors, and Non-Cooperative Object Detection

6.1 Cooperative surveillance: what ADS-B provides and what it does not

ADS-B broadcasts GNSS-derived aircraft state automatically. ADS-B Out transmits position, velocity and identity about once per second over 1090ES or 978 MHz UAT; ADS-B In receives traffic from other aircraft and ground services, adding traffic information, flight information and relay services. Compared with conventional secondary surveillance radar, reported figures give ADS-B a 0.5-second update interval against a 4–12 second radar band, and roughly 30 metres of position accuracy against roughly 400 metres. Figures 6 and 7 present those two comparisons.

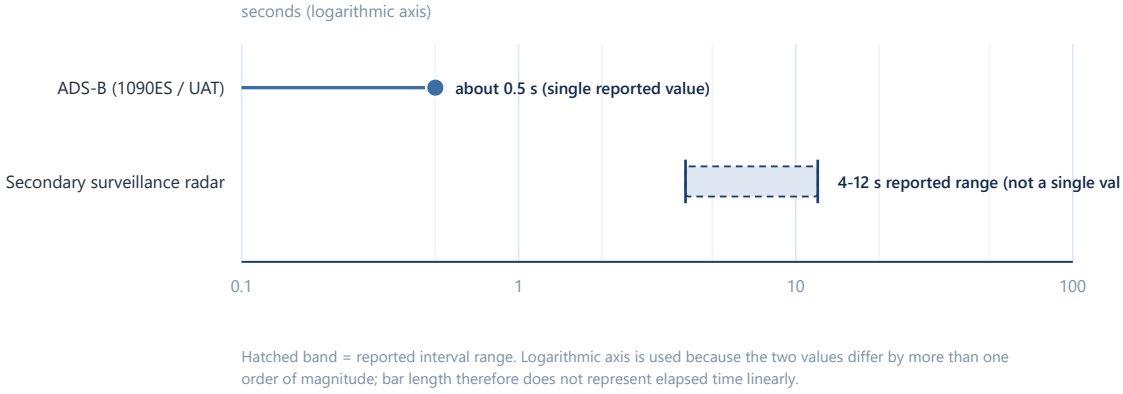


Figure 6. Reported surveillance update intervals: ADS-B and secondary radar

ADS-B is reported at about 0.5 seconds (a single reported value, plotted as a point). Secondary surveillance radar is reported as a 4 to 12 second range and is drawn as a hatched band spanning that range, not as a single value. The horizontal axis is logarithmic because the two quantities differ by more than one order of magnitude; bar and band length therefore does not represent elapsed time linearly.

Source: Reported figures cited in the supplied research-and-technology comparison and in civil aviation authority technical material. References 2 and 55 in Section 17.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

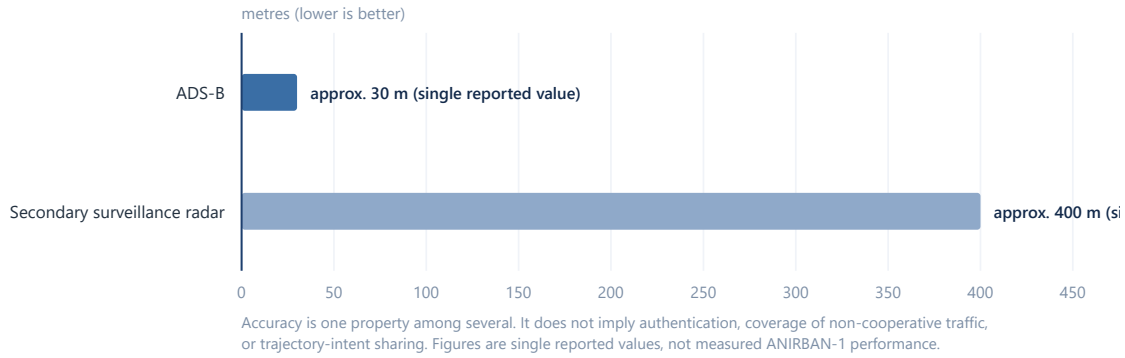


Figure 7. Reported position accuracy: ADS-B and secondary radar

Reported position accuracy of approximately 30 metres for ADS-B and approximately 400 metres for secondary surveillance radar. Both are single reported values. Lower is better.

Source: Same sources as Figure 6. References 2 and 55 in Section 17.

Figures 6 and 7 can establish that ADS-B offers more frequent and more precise cooperative state than conventional secondary radar. They cannot establish anything about ANIRBAN-1 performance. Four properties are frequently conflated and are kept separate throughout this report:

- **Update frequency** — how often state is refreshed (Figure 6).
- **Positional accuracy** — how close a reported position is to truth (Figure 7).
- **Authentication and integrity** — whether the receiver can establish origin and immutability. ADS-B was designed without security mechanisms and is susceptible to message injection and sequencing manipulation; neither figure addresses this at all.
- **Traffic coverage and intent** — ADS-B sees transponder-equipped aircraft only, and it publishes current state, not a negotiated future four-dimensional trajectory. This is precisely the gap the ANIRBAN-1 intent layer proposes to fill.

ANIRBAN-1 therefore inherits both the coverage boundary and the integrity gap of its cooperative feed. Every ADS-B-derived track enters the system with age, source and quality flags (SYS-002) and is treated as a time-bounded claim to be cross-checked, not as ground truth.

6.2 Non-cooperative sensing: four problems, not one

The supplied concept documents are internally inconsistent about sensor performance, and the inconsistency is recorded here rather than averaged away. The concept body describes a forward-scanning LiDAR at 100–300 metres, depth cameras effective to roughly 1–2 km in clear conditions, and small-target detection radars at roughly 5–10 km; the comparison table at the end of the same document lists LiDAR at 0.5–1 km and cameras at 1–5 km. These ranges differ by multiples within one document. They are carried as unverified design assumptions requiring measurement, not as specifications.

Table 7. Sensing modality analysis: what each modality observes and what limits it

Modality	What it observes	Physical and environmental limits	Integration burden	Version 0.1 role

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Modality	What it observes	Physical and environmental limits	Integration burden	Version 0.1 role
LiDAR	Three-dimensional range and local geometry as a point cloud	Rain and fog attenuate laser return; narrow field of regard; line of sight required; range figures inconsistent across sources	Several kilograms; tens of watts; heated external windows	Recorded-data research module
EO / IR camera	Pixel-level imagery and apparent motion	Requires daylight or active infrared illumination at night; degraded by fog, rain, clutter and low contrast	Roughly 1–2 kg; high data rate requiring GPU-class processing	Recorded-data research module
Airborne radar	Range, radial velocity and coarse azimuth	Coarse angular resolution; clutter and sidelobe effects; the declared range bounds what can be assured	Tens of kilograms; rotating or phased-array aperture; radome integration	Case-specific option
Cooperative surveillance (ADS-B, Mode S)	Broadcast identity and GNSS-derived state	Transponder-equipped traffic only; no native authentication	Existing certified equipment	Baseline input
DAIDALUS-style logic	Well-clear prediction derived from tracks	Entirely dependent on input track quality and the declared criterion	Software only	Benchmark, not a sensor

The detect-and-avoid literature is explicit that no single modality covers the threat set, and that fusion does not automatically produce reliable detection. ACAS Xu was designed to accept flexible sensor inputs including radar so that advice can degrade gracefully when non-cooperative sensing is absent; DAIDALUS supports multi-sensor input through configuration rather than claiming universal coverage.

Detection, identification, tracking and threat assessment are four distinct problems with four distinct error distributions. Published human-in-the-loop research makes the gap concrete: pilots lost well-clear against non-cooperative intruders significantly more often than against cooperative ones, driven by the radar's declared range and sensor noise — that is, by the distance between an instantaneous detection and a usable track with bounded uncertainty. The same body of work records that the format of an alert changes pilot response behaviour and requires dedicated training. ANIRBAN-1 therefore maintains vocabulary discipline: a LiDAR point is not a track; a track is not an identification; an identification is not a threat assessment; and a threat assessment is not a validated avoidance recommendation.

6.3 Recommended initial sensing configuration

The smallest sensible initial configuration keeps sensing entirely out of the live loop. The optional sensor module consumes recorded imagery or simulated targets, produces bounded target tracks with explicit confidence, and reports a measured detection and false-alarm distribution against a declared target envelope defined by size class, closure rate, aspect and weather categories.

Features postponed until a separate study justifies them: airborne LiDAR hardware selection; nose or belly fairing mounting with de-icing provisions, which are airframe-specific engineering decisions rather than concept-level ones; weather-radar-adjacent modes; and any universal detection claim. This report states plainly that nothing in the current evidence base supports asserting that a sensor suite detects every bird, balloon, drone, glider or non-cooperative aircraft, and ANIRBAN-1 makes no such claim. Published assessments also characterize the certification basis for non-cooperative sensors as a current gap, which limits beyond-visual-line-of-sight approval in mixed airspace.

7 Communications, Message Formats, Coordination, and Failure Behaviour

7.1 Link options compared

The original concept treated an aircraft mesh as an open frontier and a dedicated 6G network as an eventual substrate. The evidence requires both framings to be revised: IEEE 1920.1 defines infrastructure-free air-to-air ad hoc networking for civil and commercial manned and unmanned aircraft, with a network services architecture, a security framework and data models, published as a draft standard in October 2022. The design consequence is to stop inventing a transport layer and to place the research contribution above it.

Table 8. Communication options compared against the properties that matter for safety messaging

Option	Coverage	Delay and delay variation	Loss behaviour	Security	Maturity for aviation safety
Simulated / virtual network	Full control by design	Configurable	Injected deliberately	Application-level	**Chosen for Version 0.1**
IEEE 1920.1-class aerial ad hoc networking	Air-to-air, infrastructure-free	Multi-hop dependent	Topology churn dominates	Security framework defined in the standard	Draft standard (October 2022)
Existing links: ADS-B extensions, VDL-2 / CPDLC, LDACS, satcom	Operational footprints	0.5–1 s class	Engineered and regulated	Varies; LDACS designs include national cryptographic protection with TESLA-class control-data authentication	Interim field path; requires standards work
Laboratory software-defined radios	Short-range line-of-sight tests	Configurable	Measurable	Application-level	Only under authorization
Dedicated 6G aviation mesh	IMT-2030 study phase; three-dimensional coverage targets described from 120 m to 3,000 m	Targeted, not demonstrated	Study phase	Aviation profile undefined	**Future research, not a prerequisite**

No custom transmission on aeronautical frequencies is prescribed or authorized by this report. Spectrum and authorization constraints are why the virtual-network-first staging is retained.

The safety lesson from airborne-network research is consistent across three strands of published work: backbone-style architectures that keep a connected three-dimensional corridor; aviation ad hoc networks supporting direct, relay and opportunistic transmission to extend air-to-ground access into oceanic and remote airspace; and connectivity-aware mobility models that trade area coverage against network connectivity. With nodes moving at hundreds of kilometres per hour, **link intermittency and message age — not bandwidth — are the governing constraints**. Message age is therefore a first-class protocol field, every peer interaction has a declared deadline, and the safety logic remains functional when the multi-hop path fails.

7.2 Conceptual message schema

Five message types carry the safety-relevant exchange. Freshness, versioning and provenance are protocol fields, not afterthoughts.

Table 9. Conceptual message types, required fields, and semantic rules

Message	Required conceptual fields	Rules
STATE	protocol_version; sender_id; sequence; generated_at; expires_at; position; altitude; velocity; track or heading; vertical rate; uncertainty; source_quality; integrity and authentication metadata	Must be time-stamped, expiring, replay-aware, versioned, unit-defined and quality-marked
INTENT	sender_id; intent_version; valid_from; valid_until; path segments or waypoints; expected speed and altitude profile where provided; uncertainty or confidence; source	Intent is a claim about planned behaviour, not a guarantee. Revisions invalidate older versions
PROPOSAL	proposal_id; participants and tracks; candidate identifier; intended trajectory summary; assumptions; created_at; expiry; safety-check status; version	A proposal is not a command and not a clearance
ACK / REJECT	proposal_id; sender_id; outcome; reason; timestamp; expiry	Missing, late or conflicting acknowledgement means no confirmed agreement
HEALTH	component status; degraded reason; link state; current software and configuration version	Diagnostic only; non-critical telemetry must never block the prediction loop

Encoding choice (a compact binary serialization such as protobuf or ASN.1) is deferred to the prototype; neither choice alters the semantics above.

> **What the message format does not guarantee.** The integrity and authentication metadata field, when populated with a working scheme, establishes message origin and in-flight immutability. It does not establish that the sender's navigation solution is honest, that its clock is synchronized, or that its declared intent will be flown. The schema's answer to that limit is the uncertainty and source-quality fields combined with expiry: every receiver treats message content as a time-bounded claim subject to plausibility and cross-source checks. Authentication and message integrity must not be confused with verification that the underlying sensor information is truthful.

7.3 Coordination: deterministic negotiation, not consensus

The original concept invoked a blockchain-inspired consensus protocol to agree conflict-free plans atomically. That design is removed from the time-critical loop for reasons of latency, failure behaviour and, above all, evidential value.

Table 10. Coordination mechanisms compared for the conflict-resolution loop

Criterion	Timeout-bounded proposal / ACK (adopted)	Distributed consensus (Raft or PBFT class)	General-purpose blockchain ledger
Latency behaviour	Bounded by a declared deadline; the cycle continues regardless	Round-dependent; leader or partition delays unbounded in the worst case	Commit times incompatible with a seconds-scale safety cycle

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Criterion	Timeout-bounded proposal / ACK (adopted)	Distributed consensus (Raft or PBFT class)	General-purpose blockchain ledger
Outcome under peer disagreement	Deterministic tie-break rule, or suppression — a declared state	Consensus may stall; livelock handling is the hard case	Same stall risk, plus non-deterministic fork resolution
Behaviour under partition	Local detection and advisory continue; the proposal is marked unconfirmed	Majority quorum may be unreachable precisely when topology is churning	Same, with added replay and ordering exposure
What it proves about traffic truth	Nothing beyond message-level controls; plausibility checks remain required	Quorum agreement on received messages, not sensor truth	Record integrity of the ledger, explicitly not source truth
Verification burden	Small state machine; every transition testable by fault injection	Formal treatments exist but are heavyweight for a research prototype	Assurance argument effectively unavailable at safety relevance
Research value	The fault semantics *are* the candidate contribution	No marginal value over the adopted option for local conflict work	None demonstrated for time-critical collision avoidance

A ledger retains a legitimate role in ground-side air-traffic-management record integrity, as explored in the literature. It has no role in the airborne detection loop of Version 0.1.

7.4 Loss, partition, disagreement and degradation

No aircraft may hold its local conflict detection, its existing safety functions, or its degraded-mode display in reserve behind a global-consensus step. Coordination is enrichment, never a dependency. Resilience design starts from published network behaviour rather than optimism: predictive tooling can forecast data-link interruption minutes ahead, which is useful for strategic deconfliction and useless for an encounter that develops inside a minute; and complex-network vulnerability analysis of aviation systems shows that targeted attacks degrade such networks far more severely than random failures, meaning an intentional adversary will seek the connectivity-critical nodes — relay positions, timing sources and certificate services. Table 11 states the required behaviour.

Table 11. Failure and degradation behaviour: condition, required response, and evidence

Condition	Expected prototype response	Logging
Source invalid or out of range	Reject the datum; flag the source as degraded; update track status	Source, rule, timestamp and rejection reason
Track stale or expired	Mark stale; never treat the last position as current; suppress if the remaining picture is inadequate	Age, thresholds from configuration, affected outputs
Intent version changes	Invalidate the older intent; recompute the trajectory and any proposal depending on it	Old and new version, impacted candidate
No network response	Continue local prediction; mark any distributed plan unconfirmed; never assume acceptance	Timeout, peer identity, retry count, degraded state
Two proposals conflict	Apply the documented deterministic tie-break rule, or suppress the recommendation	Conflict set, rule applied, selected or none result

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Condition	Expected prototype response	Logging
Safety monitor rejects a candidate	Not shown as recommended; rejection may be made visible without implying safety	Failed checks and rule identifiers
Compute overload or deadline miss	Enter the degraded state; suppress time-expired output; watchdog-controlled recovery	Timing, component health and recovery event
Active TCAS/ACAS resolution advisory reported through a permitted input	Issue no contradictory ANIRBAN manoeuvre recommendation; apply the defined caution and mode policy	Input status, suppression reason, version

The invariant tying Sections 7.2 to 7.4 together: an aircraft's safety-relevant awareness must degrade along declared, observable states, and no transition may silently depend on reaching peers. This is what reduced dependency on continuous connectivity means once stripped of optimistic phrasing — the local loop is self-sufficient and the network is optional enrichment.

8 Onboard Software Modules and Processing Workflow

8.1 Modules, responsibilities and degraded behaviour

Eleven modules define the software boundary. The discipline that matters is that every failure mode produces an explicit state rather than silence.

Table 12. Software modules M1–M11: responsibilities and failure behaviour

Module	Responsibilities	Failure / degraded behaviour
M1 Source adapters	Decode simulator or authorized inputs; map units and coordinate frames	Reject unknown schema, impossible values, invalid status or unsupported units
M2 Validation and time	Check age, ordering, source status, sequence, clock quality and plausibility	Mark data invalid or stale; never silently reuse old state as current
M3 Track manager	Create, update, coast and expire tracks; associate duplicate observations; preserve source lineage	Tracks become degraded or expire; association ambiguity lowers confidence
M4 Intent manager	Store trajectory segments with version, sender, start time, expiry and stated limitations	Missing intent triggers an explicitly declared fallback model, not an inferred exact future path
M5 Prediction engine	Propagate candidate future positions with model assumptions and uncertainty	Widen uncertainty or report unknown when model or input quality is inadequate
M6 Conflict detector	Evaluate predicted trajectories against the configured conflict or well-clear criterion	Return detection state, time window and contributing tracks
M7 Candidate generator	Generate bounded alternative paths for evaluation; never command execution	No candidate means the system shows no reliable recommendation
M8 Coordination service	Exchange state, intent and proposal messages; correlate acknowledgements; apply timeout and deterministic tie-break policy	No complete agreement by deadline means the proposal is unconfirmed; peer acceptance is never assumed
M9 Independent proposal monitor	Re-check data age, constraints, all known traffic, proposal expiry and contradictory state	Reject or suppress the candidate with a clear reason

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Module	Responsibilities	Failure / degraded behaviour
M10 Advisory manager	Choose display mode; suppress stale advice; expose reason, quality and time horizon	Transition to a degraded or unavailable display state
M11 Health and logging	Watchdogs, processor load, dropped frames, exceptions, event logs, build and configuration hashes	Fault latched and visible; controlled restart where appropriate

8.2 Data structures

Every structure is versioned, and the message-schema evolution rules of Section 7.2 apply at the external boundary. Undocumented ad-hoc object passing between modules is prohibited: the schema is the interface.

Table 13. Core data structures and their producers

Structure	Key fields	Producer	Consumers
ObservationEvent	source_id, rx_time, schema_version, payload, quality_flag	M1 adapters	M2 validator
Track	track_id, reduced state vector, covariance or uncertainty, age, source lineage, lifecycle state (fresh, coasting, expired), association set	M3 track manager	M5, M6, M9
IntentRecord	sender_id, intent_version, valid_from and valid_until, segment list, stated assumptions, confidence	M4 intent manager	M5, M9
TrajectoryHypothesis	track_id, model tag, time-indexed states, uncertainty envelope	M5 predictor	M6, M7
ConflictSet	conflict_id, member tracks, interval, criterion tag, reason codes	M6 detector	M7, M9, M10
CandidateProposal	proposal_id, ownship track, adjustment set, constraint report, deviation cost, expiry, safety status	M7 generator	M8, M9, M10
AdvisoryState	mode (monitoring, advisory, degraded, unavailable), displayed content, quality banner, coordination status	M10 advisory manager	Display, M11 log

8.3 Deterministic core with machine learning at arm's length

Four technology routes were compared before selection. Deterministic algorithms — constant-velocity extrapolation and geometric separation tests — are predictable and easy to test, but under-report uncertainty and miss manoeuvring traffic. Optimization methods, such as polynomial trajectory families minimizing deviation cost, produce explicit trade-off logic and reproducible optima but grow combinatorially with participant count. Rule and lookup approaches demonstrate that even learned policy content can be frozen into auditable artifacts at deployment: DAIDALUS-style guidance bands are formally specified, and ACAS X uses machine-learning-optimized numerical lookup tables evaluated offline. End-to-end learning, such as deep reinforcement learning for manoeuvre generation, adapts to uncertainty patterns, but current verification methods for neural components do not meet the predictability required of a safety-adjacent gate.

The adopted architecture therefore uses a deterministic core that owns parsing, units, timeouts, state transitions, conflict thresholds and output gating. A separately monitored machine-learning inference service may contribute estimates — trajectory priors, detection confidence — only through explicit

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

interfaces and always behind deterministic checks. Technology was selected for verification tractability, not for novelty appeal.

8.4 Prediction-to-advisory workflow

The reproducible core loop is the artefact that makes the research question testable. Note the two rejection paths: the monitor gate and the explicit no-reliable-recommendation state.

```
loop every cycle T (configurable; study default 1 s):
  events   = M1.read_raw()                        # timestamped source records
  valid    = M2.filter(events, age_rule, schema_rule, plausibility_rule)
  tracks   = M3.update(valid)                     # create / associate / coast / expire
  intents  = M4.current(versioned)                # declared intent, or FALLBACK tag
  hypo     = M5.predict(tracks, intents, horizon H, uncertainty_model)
  conflict = M6.detect(hypo, criterion = C)
  if conflict:
      cands = M7.generate(conflict, constraints, ownship_envelope)
      cands = M7.evaluate(cands, tracks.all())    # ALL represented traffic, not just th
      coord = M8.propose(cands, peers, deadline D)   # ACK / REJECT / NO-RESPONSE
      for c in cands:
          gate = M9.recheck(c, tracks.age(), C, coord.status)   # independent process
          if gate == ACCEPT: M10.show(c, quality = coord.status)
          else:              M10.suppress(c, reason = gate)
      if no candidate accepted:
          M10.show(NO_RELIABLE_RECOMMENDATION)    # never the least-bad unsafe option
  else:
      M10.show(MONITORING)
  M11.log(cycle_inputs, versions, outputs, timings)   # reproducibility record (SYS-010)
```

Illustrative labelled simulation example

The following is a labelled simulation intended to clarify mechanism. It is not performance evidence and asserts no lead-time capability. Two aircraft on a crossing geometry approach a common waypoint; ownship declares a five-minute intent horizon. A constant-velocity baseline flags the conflict using an instantaneous proximity test. The intent-aware variant, propagating both aircraft's speed management against required arrival times, flags earlier but with a widening uncertainty envelope. When the peer's acknowledgement times out at deadline D, the monitor re-checks the candidate against the now-coasting track, finds that the peer's uncertainty is no longer bounded by its declared intent, and suppresses the previously accepted proposal. The display transitions from advisory to degraded with a stated reason. That transition — not a lead-time number — is the research output the scenario demonstrates.

9 Human Factors and the Advisory Interface

9.1 Design question

The design question is not whether a glowing predicted path looks impressive, but whether a crew can determine at a glance what is measured, what is predicted, what is proposed, and what is broken. Published human-factors evidence constrains the answer. Cockpit traffic display interface requirements already exist as a standardized human-interface specification covering visual and aural alert

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

characteristics and traffic display requirements. Alerting research shows measurable behavioural trade-offs: as false-alarm rates rise, pilot compliance with alerts falls, while concurrent-task performance can improve because fewer interruptions occur. Three-tier alerting showed no significant advantage over two-tier designs. Automation-assisted flying produced lower situational awareness than manual or interactive modes. Research on the ACAS X family found that the dimension and presentation of an advisory affect pilot response time, and that differing advisory formats require dedicated training. For remote operations, colour and motion cues together with labelled speed data reduce time-to-contact judgement bias, and optimized head-up display presentation reallocates attention usefully in low-visibility conditions.

9.2 Advisory display requirements for Version 0.1

Table 14. Advisory display elements: what must be visible and what must be avoided

Display element	Must be visible	Avoid
Conflict cue	Which tracks are involved, the predicted time window, and the reason for concern	A vague red icon with no explanation
Data quality	Age, source status, uncertainty, or an explicit unknown or degraded label	False precision, or a confident line derived from stale data
Candidate path	Clearly marked as proposed; expiry and version; the constraints considered	Presenting a proposal as an ATC clearance or a guaranteed safe trajectory
Coordination status	Unconfirmed, acknowledged, timed out, or inconsistent	Calling a proposal agreed when acknowledgement messages are missing
System mode	Monitoring, advisory, degraded, or unavailable	Silently hiding failure or stale inputs
Existing alerts	No contradictory ANIRBAN command; display policy designed around approved procedures	Competing alert hierarchies, or suppressing ACAS/TCAS information

9.3 The ghost path, demoted to a research visualization

The original concept described an augmented-reality windshield overlay in which the negotiated trajectory is drawn as a glowing line that the pilot follows by default, with divergence triggering alarms. That design is postponed. Version 0.1 uses a conventional advisory panel in which the proposed path is a clearly dashed line that must not resemble a flight-director command, and divergence from an advisory is an ordinary pilot act rather than an alarm condition.

The evidence in Section 9.1 predicts why. A confident overlay fed by uncertain data is the configuration that manufactures automation bias, and the compliance penalty for false alarms is measured rather than theoretical. Progression to an augmented-reality windshield is gated on controlled human-factors study results (stage V4, Section 12), because display and automation choices demonstrably change response behaviour. The first prototype displays advisory information only; separate evidence and approvals would be required before any active flight-control integration could be considered.

10 Preliminary Hazard Analysis, Safety Considerations, and Cybersecurity

10.1 Preliminary hazard register

Fourteen hazards were identified by merging the master requirement list with the supplied architecture document's HZ-01 to HZ-10 register. Severity is kept separate from probability throughout. No numeric risk scores are invented, and hazard classification authority remains a formal functional hazard assessment performed under a real certification basis by qualified personnel once the intended function and architecture are defined.

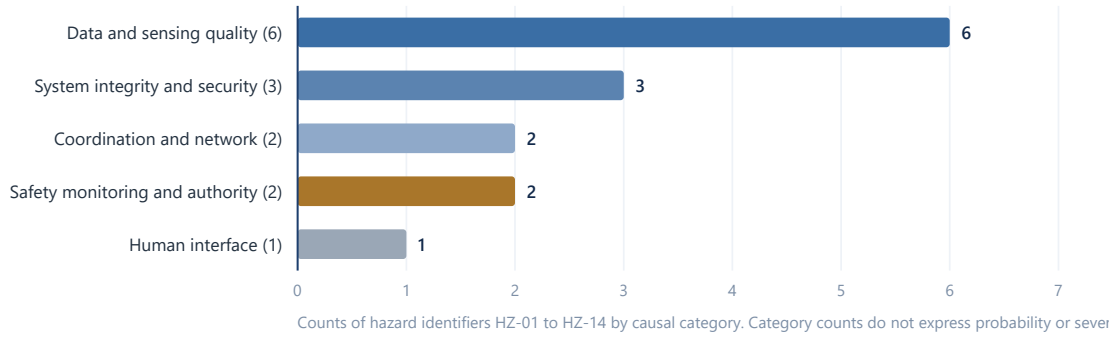


Figure 8. Preliminary hazard register HZ-01 to HZ-14 by causal category

Counts of the fourteen recorded hazard identifiers grouped by causal category: data and sensing quality 6; system integrity and security 3; coordination and network 2; safety monitoring and authority 2; human interface 1. Category totals agree with Table 15.

Source: Derived from the hazard register in Table 15 of this report; original identifiers from the supplied system-architecture design document (Rev 0.1).

Figure 8 shows where recorded hazards concentrate, and the distribution is informative about the system rather than about risk magnitude. Six of fourteen hazards arise from data and sensing quality — stale state, incorrect prediction, overconfidence, faulty fusion, missed conflicts and loss of navigation or timing. That concentration is a direct consequence of the architecture: ANIRBAN-1 adds no new actuation, so it cannot create hazards through control output, and nearly all of its failure potential lives in the quality of what it believes about the traffic picture. Category counts must not be read as probability, severity, or quantitative risk.

Table 15. Preliminary hazard register (advisory-only prototype)

ID	Hazard / cause	Potential effect	Detection	Mitigation	Validation evidence needed
HZ-01	Stale or invalid aircraft-state information	False or missed conflict prediction	Age checks, plausibility rules	Source and time validation; uncertainty and age display	Inject stale and plausible-but-wrong values; measure output gating
HZ-02	Unsafe candidate trajectory passes the planner	Advisory misleads the operator	Independent re-check at L5	All-track re-evaluation; constraint checks	Negative and boundary test cases

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

ID	Hazard / cause	Potential effect	Detection	Mitigation	Validation evidence needed
HZ-03	Communication failure or network partition; delayed intent	Peers act on inconsistent plans	Link health, timeout state	No consensus dependency; local operation; proposal expiry	Partition fault-injection reports
HZ-04	Malicious or misleading message (spoofed sender)	Coordinated advice built on a false track	Cross-source consistency, quality flags	Authentication plus explicit trust limits	Spoofing-like simulation
HZ-05	Prediction overconfidence or incorrect prediction	Display implies unjustified certainty	Calibration metrics	Uncertainty visible; suppression when evidence is insufficient	Calibration study against ground truth
HZ-06	Advisory contradicts a TCAS/ACAS alert or ATC instruction	Competing guidance confuses the crew	Resolution-advisory status via a permitted input	Suppress the conflicting proposal; document the authority hierarchy	Human-factors and procedure review
HZ-07	Timing or processor failure	Stale advisory persists; an update is missed	Watchdogs, deadline monitors	Expiry, visible degraded state, controlled restart	Fault-injection and timing reports
HZ-08	Faulty sensor fusion or false track	An object is missed or falsely declared	Confidence, modality cross-check	Bounded sensor scope; measured curves	Detection and false-alarm curves per target envelope
HZ-09	Misleading cockpit presentation or ambiguity	Operator misinterprets a suggestion	Usability trials	Clear labelling; uncertainty and status exposure	Stage V4 study data
HZ-10	Unauthorized data write path	Unvalidated data influences the system or the aircraft	Interface audit	Strict boundary; least privilege; no control outputs in the prototype	Architecture and code audit evidence
HZ-11	Missed conflict	No advisory as separation erodes	Ground-truth scoring	Declared criterion with horizon sweep	Recall and false-negative statistics on labelled scenarios
HZ-12	False alarm	Alert fatigue erodes compliance (a measured effect)	Scenario scoring	Threshold choice is a declared study parameter	Receiver-operating-characteristic trade curves across criteria
HZ-13	Loss of navigation or timing information	The whole picture degrades simultaneously	Clock-quality status	Fail-closed suppression; degraded mode	GNSS-denied scenario tests
HZ-14	Conflicting proposals between peers	Oscillation, or opposite manoeuvres proposed	Sequence and expiry correlation	Deterministic tie-break, or mutual suppression	Multi-peer adversarial scenario runs

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Two residual concerns are stated plainly. HZ-12: the false-alarm threshold cannot be chosen by engineering taste, because published alerting research shows that is exactly where crew tolerance is spent. HZ-14: the safest outcome under disagreement remains an open decision (Q5, Section 16), because two aircraft can each locally prefer a plan the other rejects, and the resolution rule must be written before multi-peer tests exist.

10.2 Cybersecurity and traffic-data integrity

The cooperative feeds on which ANIRBAN-1 depends are the most attackable links in the system, and this is measured, published ground rather than speculation. ADS-B was designed without cryptographic security mechanisms and is susceptible to message injection and sequencing manipulation. Published countermeasures trade cost against partial protection: a framework providing message authentication and integrity without depending on public-key cryptography at a bandwidth overhead below 50 per cent, and a planned aviation datalink security architecture combining national cryptographic algorithms for confidentiality and integrity with TESLA-based authentication of control data at an average byte overhead below 10 per cent. The operational threat is documented: national security authorities have reported foreign intelligence penetration of connected drone inspection platforms, with exploited weaknesses catalogued as unpatched vulnerabilities and weak credentials, ambiguous permission boundaries, unencrypted data transmission and device-firmware backdoors. Regulatory response is moving from guidance toward mandate, with a mandatory national standard on civil-UAV data-link cybersecurity in drafting that covers identity authentication, anti-forgery and anti-tampering, secure updates and anti-replay protection.

Table 16. Threat, design control, and what each control does not prove

Threat or fault	Design control	What the control does not prove
Spoofed sender or forged message	Identity and key management; message authentication where suitable	Does not prove that the sender's sensors or reported intent are truthful
Replay or stale message	Sequence numbers, timestamps, expiry, anti-replay checks, clock-quality status	Does not eliminate all time-source failure or compromised-sender risk
Corrupted or malformed data	Strict schema validation, range checks, version negotiation, fail-closed handling	Plausible values can still be wrong
Network partition or jamming	Timeouts, link health, bounded retries, degraded mode, local prediction	Cannot guarantee message availability when the link is absent
Compromised process	Process isolation, least privilege, signed builds and updates, logging, watchdogs	Requires a broader system threat model and verification
Unsafe configuration or update	Versioned configuration, integrity checks, controlled installation, rollback	Does not replace authorized maintenance and configuration management

Privilege separation is the architectural counterpart: the network gateway, prediction service, user interface and maintenance or log-export path run under separate privileges, and no ground analytics service or internet-reachable component may write an unchecked trajectory into the advisory path.

10.3 Authority boundaries and operational limitations

The pilot and air traffic control retain authority. ANIRBAN-1 output is explicitly advisory and does not change the legal or operational authority structure. Liability questions remain open and are noted rather than resolved: in concept the crew retains final authority while the system negotiates with other aircraft, and any incident investigation would examine algorithmic decisions, which is why validation evidence and complete logging (SYS-010) are treated as first-class deliverables.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Operational limitations are stated without softening. Full benefit requires equipage density, so legacy unequipped traffic does not benefit. Non-cooperative obstacle detection is limited by weather, lighting, geometry and processing. GNSS-denied conditions degrade intent sharing. Fallback to existing ATC separation and to TCAS/ACAS is therefore mandatory, not optional.

11 Applicable Standards and Potential Certification Pathway

The standard inventory in the supplied documents required correction before use, because several citations name superseded revisions or imply a compliance relationship that does not exist. The system-development guidance cited as ARP4754A has been succeeded in the development-process chain by ARP4754B, which is aligned with ARP4761A for safety assessment; ARP4761A provides the hazard-identification and risk-assessment methods (functional, preliminary system and system safety assessment). DO-178C remains the primary airborne-software assurance means with development assurance levels A to E, and DO-254 is the matching process for complex electronic hardware. The functional relationship matters more than the catalogue: ARP4754-class guidance ensures the system is developed correctly, ARP4761-class guidance ensures the correct system is developed, and the outputs of the development process — safety requirements and assurance-level allocation — drive the software and hardware verification activities beneath them.

Table 17. Standards and guidance mapped to ANIRBAN-1 activities

Domain	Guidance (current designation)	Role for this programme	Status caveat
Airborne software	DO-178C / ED-12C	Requirements-to-code verification rigour for the deterministic core, if an assurance track is pursued	Recognized by FAA advisory circular AC 20-115D as an acceptable means, not the only means; applicability is project-specific
Complex electronic hardware	DO-254 / ED-80	Applies to any FPGA or ASIC developed for radios or gate arrays in a future H2 or H3 build	Only relevant if complex hardware is actually developed
System development	ARP4754A, succeeded by **ARP4754B**	Development process, assurance-level allocation, requirements linkage	A process framework, not a certificate
Safety assessment	ARP4761, succeeded by **ARP4761A**	Functional, preliminary system and system safety assessment methods behind the hazard register in Table 15	A formal assessment requires a defined intended function and aircraft
Environmental and EMC	DO-160-class test categories	Qualification of airborne-equipment candidates	Applies to H3-class design studies only
Cybersecurity	DO-326A / DO-356A-class process; emerging national datalink security standards	Threat model, security requirements and verification	Rule-creation status varies by authority; current mandates must be verified per jurisdiction
Human factors	ARP5365A cockpit traffic display interface standard, plus authority human-factors policy	Display and alerting design evidence for stage V4	The standard covers traffic displays; it does not address augmented-reality overlays specifically

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Domain	Guidance (current designation)	Role for this programme	Status caveat
Modification approval	FAA and EASA supplemental type certificate routes and equivalents	Relevant only after a real aircraft modification is proposed	Listing an approval route is not approval; no timeline is asserted

> **Two statements that must travel with any excerpt of this section.** First, no standard named in this report means ANIRBAN-1 currently complies with that standard. Second, certification is not a final paperwork phase: it shapes system requirements, architectural independence, test evidence, configuration control and human-factors work from the beginning. Assurance levels are not assigned in this document; they are outputs of a formal assessment under a defined certification basis, not inputs to a concept.

The credible pathway from software simulation to hardware-in-the-loop testing, controlled research equipment and potential aircraft integration is staged in Section 13. Stage V5 exists specifically to convert standards awareness into an aircraft-specific feasibility decision made with a manufacturer or design organization and the applicable authority; stage V6 admits trials only under proper authorization, initially passive and record-only.

12 Validation Methodology, Baselines, Metrics, and Acceptance Criteria

12.1 Validation stages

No live aircraft installation is needed to establish the first meaningful technical result. The programme advances through declared stages, each with exit evidence rather than a date.

Table 18. Validation stages V0–V6 with exit evidence

Stage	Environment	Test focus	Exit evidence
V0 Model review	Requirements and code review	System boundary, definitions, assumptions, data contracts, threat model	Approved review checklist; open issues recorded
V1 Software simulation	Multi-aircraft simulator or custom scenario engine	Crossing, converging and overtaking geometries; intent changes; uncertainty; multiple traffic	Reproducible scenario set; baseline metrics
V2 Network simulation	Virtual network with injected loss, latency, reordering, duplication, partition and replay-like cases	Message semantics, timeouts, inconsistent intent, degraded state	Fault-injection results; safe output-gating evidence
V3 Hardware-in-the-loop	Bench computer with simulated I/O; optional lab sensors or radios	Timing, restarts, resource exhaustion, interface fault behaviour	Timing and recovery reports; confirmation that no unauthorized radio-frequency transmission occurred

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Stage	Environment	Test focus	Exit evidence
V4 Human factors	Mock display and controlled operator study	Comprehension, response, workload, alert interpretation, trust calibration, rejection of incorrect advice	Study protocol, ethics handling where applicable, data and analysis
V5 Aircraft feasibility study	Aircraft-specific design and safety review before installation	Interfaces, power, EMC, environmental conditions, airworthiness and certification route	Qualified engineering decision and approval plan
V6 Approved ground and flight trials	Only under proper authorization and an approved test plan	Incremental passive and record-only operation first, then carefully bounded advisory trials	Formal test reports; authority and manufacturer approvals

Retained from the supplied architecture document. BlueSky is one candidate open-source air-traffic simulation starting point for stages V1 and V2.

12.2 Baselines and the ablation discipline

Three baselines define what "improvement" would mean. **Baseline A** is constant-velocity extrapolation with a geometric proximity test. **Baseline B** is a DAIDALUS-style detect-and-avoid implementation: NASA's reference logic provides a well-clear definition, look-ahead prediction of violation intervals, conflict and recovery guidance bands, and graduated alerting, and is openly available with prototype implementations and formal specifications. **Baseline C** is a published four-dimensional trajectory conflict-detection and resolution method whose documentation permits reproduction; polynomial-based distributed resolution minimizing deviation from original trajectories is the candidate recorded in the supplied comparison.

The ANIRBAN variant is intent sharing plus uncertainty handling plus fault-aware coordination. **Ablation runs must disable one feature at a time.** Without ablation, a measured improvement cannot be attributed to any component — which is precisely the standard that concept claims of this kind have so far failed to meet.

12.3 Metric families

Table 19. Metric families to be defined before testing

Metric family	Candidate measures
Detection	Conflict detection rate, missed-conflict rate, false-alarm rate, warning lead-time distribution
Prediction	Position prediction error by horizon, calibration of uncertainty, behaviour under intent change
Resolution	Feasibility rate, modelled constraint violations, path deviation, manoeuvre feasibility, oscillation and repeated replanning
Network	Message age, end-to-end delay distribution, packet loss, stale-message rejection, recovery after partition
Runtime	Processing-time distribution, deadline misses, processor and memory use, restart time, event-log completeness

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Metric family	Candidate measures
Human factors	Time to understand an advisory, correct manoeuvre selection, workload, trust calibration, and rates of inappropriate acceptance or rejection

Acceptance thresholds are set before testing, derived from stated research aims and engineering constraints. Arbitrary percentages are not borrowed, and no flight-safety assurance is claimed from small simulation studies.

12.4 Research questions that define success

1. Does sharing a planned trajectory improve prediction lead time over current-state-only traffic, measured against Baselines A to C?
2. Under what message-loss conditions does distributed coordination stop adding value?
3. Does uncertainty-aware sensing reduce missed detections without creating unacceptable false alerts, judged by measured detection and false-alarm distributions?
4. Does the interface help operators correctly distinguish safe from unsafe proposals?
5. What measurable benefit survives after accounting for false alarms and for legacy or unequipped aircraft?

13 Development Roadmap and Evidence Gates

The original concept committed to a fixed five-to-seven year schedule with calendar milestones. That commitment is replaced by phase gates whose exit criteria are evidence, because certification duration depends on the certification basis, the authority and the programme scope, none of which can be asserted from a concept document. Figure 9 shows the illustrative sequencing; the gates govern.

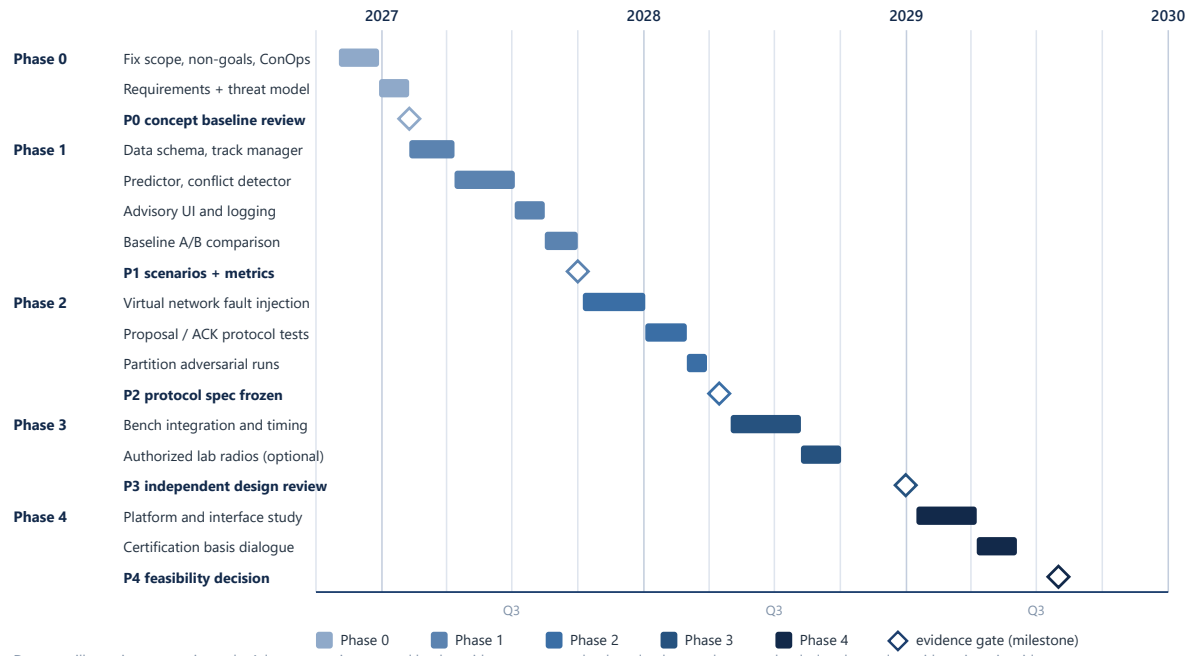


Figure 9. Development roadmap: illustrative sequencing with mandatory evidence gates

Five phases from concept baseline to research aviation study. Bars show illustrative activity windows; open diamonds mark evidence gates (P0 concept baseline review, P1 repeatable scenarios and baseline metrics, P2 protocol specification frozen, P3 independent design review, P4 feasibility decision). Dates are planning assumptions, not commitments.

Source: Phase content from the supplied system-architecture design document (Rev 0.1). Calendar dates are illustrative planning assumptions introduced in this revision and are not derived from any source estimate.

Reading the schedule correctly. The visual schedule is subordinate to the technical decision gates. Assigning a date does not make aircraft integration, operational trials or certification automatic: Phase 4 cannot begin without a selected platform and an engaged authority, and no trial in Phase 5 or beyond may proceed without the approvals its gate requires. A phase may not be declared complete without its exit evidence, regardless of elapsed time.

Table 20. Phase gates: scope and exit evidence

Phase	Scope	Exit gate (evidence, not calendar)
Phase 0 — Concept baseline	Intended use, airspace and user context, source interfaces, assumptions, non-goals, hazards, benchmark plan	Concept of operations; system boundary; initial requirements; literature evidence map
Phase 1 — Simulation MVP	Data schema, track manager, predictor, conflict detector, advisory interface, logging	Repeatable scenarios; baseline metrics; no radio or aircraft modification
Phase 2 — Robustness and network	Virtual network; proposal and acknowledgement protocol; timeouts; authentication simulation; partition cases	Protocol specification; threat model; failure-state evidence
Phase 3 — HIL prototype	Bench computer; simulated avionics I/O; optional controlled lab sensors or radios	Interface test report; timing and fault report; independent design review

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Phase	Scope	Exit gate (evidence, not calendar)
Phase 4 — Research aviation study	Select platform and function; engage a manufacturer or design organization and the authority	Aircraft-specific feasibility; certification-basis discussion; test plan
Phase 5 — Authorized trials	Only after required approvals; passive and record-only first, then bounded advisory operation	Conformity and test evidence required by the approved programme
Phase 6 — Operationalization study	Procedures, training, maintenance, interoperability and continuing airworthiness — only if evidence justifies it	Certification and operational evidence; no approval assumed

Phases 5 and 6 lie beyond the roadmap drawn in Figure 9 and are entered only on the strength of Phase 4 evidence.

13.1 Cost and retrofit considerations (qualitative)

Table 21. Cost and complexity categories by hardware option

Cost category	H1 (bench)	H2 (research island)	H3 (airborne-class)
Hardware	Software development time only	Workstation plus adapter and watchdog units	Qualified line-replaceable-unit development and testing
Integration	Simulated I/O	Laboratory interfaces; optional authorized radios	Airframe-specific engineering, EMC and environmental qualification
Assurance	None required (research activity)	Internal design review	Full ARP4754B and ARP4761A-class process, with DO-178C rigour per assigned assurance level
Approval	Not applicable	Not applicable	Supplemental-type-certificate-class modification approval per aircraft

The original concept report's per-aircraft dollar bands, power-draw figures, fuel-burn impact estimate and itemized vendor pricing are removed as unverified. Restating them would require vendor quotations under a defined installation scope and airframe-specific weight and power analysis, neither of which exists. Adoption incentives such as equipage mandates are policy questions noted here but not forecast.

14 Discussion, Research Gap, Proposed Contributions, and Limitations

14.1 Mapping the occupied territory

Layering ANIRBAN-1 against the evidence base produces an honest overlap map. Trajectory mathematics: occupied. Conflict screening and detection research, including space-time reachability treatments of motion and pilot-intent uncertainty, and polynomial-based distributed resolution: occupied. Manoeuvre optimization including learning-based methods: occupied. Alerting and guidance logic with formal verification: occupied by DAIDALUS, and being operationally superseded by the ACAS X family. Intent sharing to ground services and between air-traffic systems: under active deployment in national TBO programmes. Air-to-air broadcast of current state: fielded through ADS-B. Air-to-air ad hoc networking services: standardized in draft form.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

14.2 The surviving research gap

What remains genuinely open is a coherent, testable specification of fault-aware peer coordination: message expiry and staleness semantics; deterministic disagreement and tie-break outcomes; explicit unconfirmed states when peers do not answer; partition behaviour that preserves local detection; and an advisory interface that exposes all of this to a human. These are protocol-and-evidence problems rather than algorithm problems, which is why Sections 7 and 9 carry the novelty burden in this revision.

14.3 Candidate contributions

Table 22. Candidate contributions stated as hypotheses, with the evidence each requires

ID	Hypothesised contribution	Evidence required to claim it	Current status
C1	A fault-aware state and intent exchange schema with uncertainty, time validity and explicit degradation rules	Schema specification; stage V2 results showing correct expiry and staleness behaviour versus an ADS-B-style state-only feed	Specified conceptually; untested
C2	A coordination protocol that remains predictable under disagreement and message loss	Deterministic tie-break specification; adversarial multi-peer runs at stages V1 and V2; no undefined states in the state machine	Design drafted; the disagreement outcome is still an open decision (Q5)
C3	A combined reproducible evaluation of trajectory sharing, sensing uncertainty and advisory quality in one simulation and hardware-in-the-loop environment	Open scenario corpus; baseline comparison table; ablation attribution	Environment plan exists; nothing executed
C4	An advisory presentation that exposes uncertainty and data freshness rather than a confident path	Stage V4 study with calibrated-trust and correct-rejection measures, referenced against published ACAS Xu display-response findings	Design principles set; study not run

A credible novelty claim requires a documented literature search and baseline comparisons. The literature already contains multi-aircraft four-dimensional conflict detection and resolution, optical detect-and-avoid, collision avoidance optimization, and well-clear algorithms, so these four items are the defensible residue and are stated as hypotheses contingent on the paired evidence.

14.4 Limitations of this report

- **No measurements exist.** Every ANIRBAN-1 performance characteristic in this report is a design target or a hypothesis. Nothing has been flown, and no simulation campaign has been run.
- **Unverified source numbers are flagged, not corrected.** Where the supplied documents conflict with each other — sensor ranges, alert lead times, uptime targets — both values are recorded and the item is marked for author or measurement confirmation.
- **The project name has no public footprint.** Searches across academic and government platforms returned only unrelated personal-name matches for the project designation, which is consistent with a private research project and means no external corroboration of its claims is available.
- **Standards status is as-reported.** Revision designations and advisory-circular status were checked against the supplied references; a formal applicability determination belongs to the certification authority for a specific project.
- **Cost and schedule are qualitative.** No verified vendor, weight, power or approval-duration data was available, so none is asserted.

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

15 Decisions Made During Revision of the Original Concept

15.1 Disposition of the original functional elements

Twelve functional elements of the original concept were dispositioned. Figure 10 summarizes the outcome and Table 23 records the reasoning element by element. The changes matter because they define what Version 0.1 is allowed to claim: retaining intent sharing and the independent safety monitor while removing consensus and postponing control execution, augmented-reality overlays, universal sensing and 6G dependency produces a smaller programme whose research question can actually be answered with the resources a research team has.



Figure 10. Disposition of the twelve functional elements of the original concept

Three elements retained, four revised, four postponed behind evidence gates, and one removed from the time-critical safety loop. Segment widths are proportional to counts and total twelve.

Source: Derived from the element-by-element disposition recorded in Table 23 of this report.

Table 23. Element-by-element disposition and rationale

Original element	Disposition	Reason
Aircraft as connected nodes in an air-to-air network	Revised	Retain the networking concept but anchor it to IEEE 1920.1-class airborne ad hoc networking rather than an invented transport layer
Aircraft-state and planned-trajectory sharing	Retained	The core testable hypothesis; complements ADS-B state broadcast, which does not carry a negotiated future four-dimensional path
Forward-looking conflict prediction	Retained	Must be evaluated against DAIDALUS-style well-clear look-ahead rather than asserted
Multi-aircraft coordination and resolution proposals	Revised	Restricted to timeout-bounded proposal and acknowledgement exchange with explicit disagreement handling
Blockchain or PBFT-style consensus in the safety loop	Removed	Adds latency and failure modes; cannot establish sensor truth; unnecessary for local detection
Automatic trajectory execution on the autopilot	Postponed	Requires a separate flight-control integration, human-factors and certification case before any control path exists
Augmented-reality windshield ghost path with auto-follow default	Postponed	Start from a conventional advisory display; published alerting and situational-awareness findings require evidence first
LiDAR, optical and radar non-cooperative sensing	Postponed as a separate study	Retained as an offline research module; detection claims need a measured target envelope

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Original element	Disposition	Reason
Dedicated 6G aviation mesh	Postponed	Requirements work ongoing with commercial framing around 2030
Independent safety monitor	Retained	A strong element of the original concept; formalized as layer L5 with declared independence limits
Ground and ATC integration	Revised	ATC oversight retained; any operational-concept change is institutional and needs controller human-factors evaluation
Certification appendix asserting DAL A and an STC route	Revised	Converted from asserted compliance mapping into a pathway analysis tied to each standard's actual role

Category totals agree with Figure 10: three retained, four revised, four postponed, one removed.

15.2 Claims requiring confirmation, and numbers removed

Table 24. Unresolved claims and removed figures

Claim or item	Source	Gap type	Disposition and confirmation route
TCAS resolution advisory issued roughly 20–25 seconds before collision	Original concept report	External, verifiable	Retained as an unverified original claim. Public material describes a roughly five-second pilot response window on a resolution advisory; the 20–25 second lead figure needs its own citation before use
Communication uptime above 99.999 per cent and mesh latency around 100 ms	Original concept report	External, engineering	Removed as an unverified target. Requires link-level simulation and authorized radio trials
Conflict resolution success in X per cent of scenarios; no near-mid-air-collision events in thousands of flight hours	Original concept report	Placeholder, internal	Retained as placeholders. The team must set numeric acceptance thresholds before testing
Development assurance level A assigned to collision-avoidance functions	Original concept report	Regulatory	Removed. Assurance levels are outputs of a formal assessment under a defined certification basis, not concept inputs
Per-aircraft hardware cost of roughly USD 100,000 to 500,000, with itemized sensor, compute and radio prices	Original concept report	Unverified, external	Removed. Requires current vendor quotations under a defined installation scope
Five-to-seven-year certification timeline	Original concept report	Unverified, regulatory	Replaced by evidence gates (Section 13). Duration depends on basis, authority and scope
Fuel-burn impact below 1 per cent after noise-mitigation measures	Original concept report	Unverified, engineering	Removed. Requires airframe-specific weight and power analysis
Sensor range figures inconsistent between the concept body and its own comparison table	Original concept report	Internal inconsistency	Both values recorded in Table 7; flagged for measurement rather than averaged

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

Claim or item	Source	Gap type	Disposition and confirmation route
Contribution statements containing unfinished area placeholders	Contribution note	Internal, authorship	Reproduced verbatim in Section 19 with the placeholders visibly flagged as author-action items

15.3 Presentation defects corrected in this edition

- The standalone opening chart gallery was removed. Each visualization now sits beside the argument it supports and is cited explicitly in the text (Figures 1 to 10).
- The duplicated glossary and contributor block at the end of the exported PDF was eliminated; each appears once, as Sections 18 and 19.
- Tables that collapsed into single merged cells on PDF export were rebuilt with proper header groups and page-break behaviour, so headers repeat across pages.
- The radar update interval, previously drawn as a single bar at the top of its reported range, is now drawn as a hatched band spanning the reported 4–12 second range on a logarithmic axis with the scale choice explained (Figure 6).
- Generic series names were replaced with descriptive labels carrying units; redundant legends were removed; estimates, projections, ranges and programme-reported claims are labelled as such in every caption.
- Charts are now inline SVG rather than canvas rendered from remote scripts, so figures cannot appear blank or malformed in the PDF edition.

16 Conclusions and Next Steps

16.1 Conclusion

ANIRBAN-1 is worth building precisely because the systems around it already work. Probabilistic multi-directional collision avoidance, open well-clear alerting logic with formal specification, verified four-dimensional trajectory sharing, and standardized air-to-air ad hoc networking all exist. The open frontier is not smarter avoidance and not a new network. It is whether aircraft can share intent and coordinate under fault, uncertainty and disagreement without ever making their existing safety depend on that cooperation. That question is answerable with bench computers, simulators and honest baselines, and the scope defined in this report is sized to answer it.

Until the ablation baselines of Section 12 have been run, the correct statement of ANIRBAN-1 status is that it is a well-posed research hypothesis with a credible, low-cost test path — not a demonstrated safety improvement, and not a replacement for any existing system.

16.2 Open decisions gating implementation

Table 25. Open questions, why they matter, and what triggers each decision

#	Open question	Why it matters	Decision trigger
Q1	First target: airliner, research aircraft, or unmanned aircraft	Changes the traffic model, interfaces, target environment and airworthiness path	Before selecting real equipment or a certification strategy

ANIRBAN-1 Integrated Technical Concept Report - Rev 1.0

#	Open question	Why it matters	Decision trigger
Q2	Which separation or well-clear criterion governs	All conflict results depend on the chosen criterion and operating domain	Before creating test ground truth
Q3	Which traffic and intent standard is exchanged	A message format without an interoperable service is not deployable	Before integration beyond simulation
Q4	How uncertainty and unknown data are represented	Determines false negatives, false alarms and suppression behaviour	Before baseline comparison
Q5	Safest coordination outcome under disagreement	Multi-aircraft interactions can oscillate or produce contradictory advice	Before enabling peer proposal tests
Q6	Which sensor target envelope is in scope	Prevents unsupported universal-detection claims	Before selecting any LiDAR, camera or radar module
Q7	Which authority and certification basis applies	Determines evidence, rigour and allowable operation	Before any aircraft modification

16.3 Immediate next steps (twelve-month action set)

1. Freeze the Version 0.1 requirements (Table 4) and the message schema (Table 9).
2. Stand up the Baseline A and B simulation harness with the ablation switchboard described in Section 12.2.
3. Run the stage V2 fault-injection catalogue against the deterministic coordination protocol and record every state transition against Table 3.
4. Document the threat model including the explicit authentication-limits statement from Section 7.2.
5. Draft the stage V4 human-factors protocol against cockpit traffic display interface requirements.
6. Open decision records for Q1 to Q7 and complete the contributor statements in Section 19.

17 References

Supplied project materials are treated as project context, not as verified evidence of system performance. External sources were checked for status and role; where a claim could not be established it is marked for author verification in Table 24 rather than presented as confirmed.

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Verification note. A public-name check found no aviation-system usage of the project designation across academic and government platforms; only unrelated personal-name matches were returned. This is consistent with a private research project and means no external corroboration of project-specific claims is available. All ANIRBAN-1 features and expected benefits remain proposals until they are defined and measured in a controlled study.

18 Glossary

ACAS / TCAS
Airborne Collision Avoidance System / Traffic Collision Avoidance System.

ADS-B
Automatic Dependent Surveillance–Broadcast; a cooperative traffic-state source where aircraft are equipped and coverage is available.

Advisory-only
Displays information and recommendations; outputs no flight-control command.

AFDX
Avionics Full-Duplex Switched Ethernet (ARINC 664 Part 7); an aircraft-specific networking possibility, not a universal interface.

4DT
Four-dimensional trajectory: latitude, longitude, altitude and time.

ATC
Air Traffic Control.

CD&R
Conflict Detection and Resolution.

CDTI
Cockpit Display of Traffic Information.

DAL
Development Assurance Level (A to E), an output of formal safety assessment.

DAIDALUS
NASA Detect and Avoid Alerting Logic for Unmanned Systems; an open reference implementation of well-clear alerting and guidance bands.

Degraded state
The system remains available for limited purposes but cannot support its normal confidence level; advice may be suppressed.

FMS
Flight Management System.

FHA / PSSA / SSA
Functional Hazard Assessment / Preliminary System Safety Assessment / System Safety Assessment (ARP4761-class methods).

HIL
Hardware-in-the-loop: prototype hardware tested against simulated inputs, plant and traffic.

ICD

ANIRBAN-1 Integrated Technical Concept Report · Rev 1.0

Interface Control Document: message definitions, units, rates, access rights and failure semantics.

Intent
A stated intended future trajectory; a claim about planned behaviour, not proof of what the aircraft will do.

MVP / Version 0.1
Minimum viable prototype; here, a simulated, advisory-only research implementation.

RA / TA
Resolution Advisory / Traffic Advisory.

STC
Supplemental Type Certificate; an approval route for a modification to an aeronautical product.

TBO
Trajectory-Based Operations.

Unconfirmed state
A declared coordination outcome when a peer does not respond by the deadline; never treated as acceptance.

Well-clear / conflict criterion
The declared mathematical threshold used to label potentially unsafe proximity; it must be defined for the research domain.

19 Contributor Statements

> **Author action required.** The contribution statements below are reproduced verbatim from the project's own contribution note. Two statements contain unfinished area placeholders in the source document; they are flagged here rather than filled in, because no contributor's responsibility may be invented or embellished.

Abu Kayes Huda: Conceived the entire concept and designed the full system. Developed the IoT mesh architecture, node communication, real-time data routing, redundancy/failover, autonomous aircraft safety logic, and system integration.

Jaidul Islam Alvi: Participated in early discussions and provided minor feedback on [specific area] — *unfinished in the source document; to be completed by the contributor.* Reviewed draft material.

MD Sakib Hasan: Participated in discussions and provided minor input on [specific area] — *unfinished in the source document; to be completed by the contributor.* Reviewed the final report/presentation.

This revision was prepared by an aviation systems engineering and technical editing review of the supplied project materials. Corrections to the original concept are recorded in Section 15 with their rationale. Placeholders and unverified figures are deliberately left unresolved rather than completed by inference.

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